

Smallest 128Gb MLC NAND chip

SanDisk has developed the world's smallest 128 Gb NAND flash chip using its 19nm process technology
<http://goo.gl/t8C0P>

SSD future?

According to a group of researchers, SSDs have a bleak future
<http://goo.gl/c7Ju1>

SSD Test

	ADATA	Kingston	Strontium	Kingston
	ADATA S510 120 GB	Kingston SSDNow V+ 200 90 GB	Strontium Python Series 240GB	Kingston HyperX 120GB
	AS510S2-120GM-C	SVP200S3/90G	NA	SH100S3/120G
	10150	8500	21900	13500
	90.8	101.38	97.96	120.76
	55.19	51.64	68.39	64.31
	55.19	51.64	68.39	64.31
	Y	Y	Y	Y
	111.79	83.84	223.57	111.79
	Sandforce SF2281VB1	Sandforce SF2281VB1	Sandforce SF2281VB1	Sandforce SF2281VB1
	16 x Intel/Micron 29F64G08CBAAA	16 x Intel/Micron 29F64G08ACME	NA	16 x Intel/Micron 29F16B08CCME2
	SATA 3	SATA 3	SATA 3	SATA 3
	Y	Y	N	Y
	N	N	N	Y
	N	N	N	Y
	3	3	3	3
	468.9	487.5	488.8	479.2
	414.4	447	436.3	433.3
	472.6	472	482.7	475.5
	445.6	450.6	466.6	448.4
	209.6	215.1	216.6	267.5
	198.5	203.2	204.4	256.3
	148.8	121.5	251.3	166.7
	149.7	121.7	250	166.9
	204.79	200.95	212.24	264.4
	18.35	17.5	18.77	18.93
	86.83	88.15	129.86	105.46
	0.1200	0.1610	0.0630	0.0810
	142.28	115.96	235.49	158.62
	52.15	54.63	55.38	52.39
	135.28	112.28	206.76	147.42
	0.2920	0.3100	0.2780	0.2790
	396	372	551	446
	144.34	128.27	149.68	212.62
	101.79	84.96	113.71	134.28
	118.15	104.09	132.03	167.3
	4695	4604	4860	4991
	5.31	5.27	5.41	5.43
	21.61	20.02	24.71	23.11
	21.6	21.45	21.84	22.26
	8.17	8.16	8.18	8.12
	1.4	1.39	1.4	1.4
	37.56	36.26	39.79	48.98
	15.58	15.47	15.86	16.38
	189.23	185.67	245	238.97
	159.34	155.15	216.37	196.07
	156	130.73	205.57	181.07
	141.09	117.81	170.47	157.81
	87.76	81.42	109.97	101.13
	63.13	74.83	103.72	94.59
	11.68	15.35	11.26	10.23



Intel Series 520 240 GB SSD

Also the V+ 200 series uses a SATA 3 interface as opposed to the SATA 2 interface found on the V+ 100 series drives.

The main difference between the V+ 200 series and the HyperX SSDs is the use of asynchronous memory for the V+ 200. The asynchronous memory is economical as compared to the synchronous memory as it limits the data transfer speed between the controller and the flash memory chips. So even though you have the same SSD controller, the choice of flash memory chips determines the speed and ultimately cost of the SSD. The V+ 200 series is targeted at more mainstream user base. In appearance though, you will not be able to tell the V+100 apart from the V+200 series of drives.

The ADATA S510 is the younger sibling of the higher end S511 series SSDs. The major difference being that S510 sports asynchronous flash memory chips to keep the costs low. Apart from the memory type, rest of the things remain the same - SF 2281 controller and SATA 3 interface. Around 8 GB of space of the total 128 GB (16 x 8GB MLC NAND flash units) is kept aside for over provisioning for removing garbage. It comes bundled with a 2.5-inch to 3.5-

inch adapter along with some screws.

Kingmax is a relatively new player in the SSD segment as far as the Indian market is concerned. We got its SMP35 120GB SSD which is targeted at mainstream users. It uses the SF2281 controller.

Kingmax being a memory module and flash devices manufacturer, is one of the only SSD manufacturers to use their own memory - the Kingmax K1C08G-ACAMMAA, which is an asynchronous MLC NAND flash chip. It has a SATA 3 interface and it bundles in a 2.5-inch to 3.5-inch adapter along with a SATA cable and a molex to SATA power cable.

Silicon Power sent us their 120GB SSD from the Velox V30 series, which according to its cover, is an enterprise class SSD. It houses the SF 2281 controller along with 16 x 8GB of synchronous Intel NAND memory. It has a SATA 3 interface. Around 8 GB of space of the total 128 GB (16 x 8GB MLC NAND flash units) is kept aside for over provisioning for removing garbage and getting better IOPS. It comes bun-



ADATA S510 120 GB SSD